

Proposed Research: Contextuality in LLM Gender Pronoun Resolution

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Many sciences rely on the assumption that a set of latent variables can fully characterize a system [3]. Indeed, the notion that nature has "pre-prepared" a system prior to our analysis of it is central to classical mechanics. Quantum mechanics, however, required physicists to step away from this notion. Ongoing experiments related to Bell's Theorems [2] have shown that in measuring quantum states, the entire measurement scenario must be considered. In such cases, the measurement itself *cannot be disentangled from the context in which it occurred*. This is to say that, as Bell showed, if Ava and Bhavya had inconsistent measurements of the same particle, the inconsistency cannot be mapped to changes in any set of hidden local variables. When this is the case, the system is said to be "contextual."

Instead, the measurement itself is a function of the measurement scenario. A measurement scenario, as defined in [8] is a tuple $\langle \mathcal{X}, \mathcal{M}, \mathcal{O} \rangle$, where $\mathcal{X}$ is the data, $\mathcal{M}$ is the measurement cover (the set of possible contexts), and $\mathcal{O}$ is the set of observations. In [8] and related works, the authors have shown that this dependency on measurement scenario exists not only in quantum particles, but is exhibited in some instances by LLMs as well. From a linguistic standpoint, we can consider this from the point of view of linguistic underdeterminacy—a highly researched area in pragmatics and formal semantics [1, 7].

There are two approaches to ascertaining contextuality in social and behavioral settings. The first is the Sheaf-Theoretic approach [8, 9], and the second comes from Mathematical Psychology and is known as the "Contextuality-by-Default" framework [5, 4]. For settings in which measurements extend beyond binary outcomes, the latter is preferred. Given that our interest, however, is in a binary pronoun resolution task, the former may suffice. Personally, I prefer the former because it is easier to explain and also has nice connections to Categorical Grammars (for any formal linguistics who may encounter this work).

While the mathematics underpinning is quite technical, ascertaining whether a system is contextual is quite simple. One must only obtain the probabilities of each outcome in each context and determine whether or not they can be coupled to form a *valid* joint probability space. This reduces to solving a set of linear equations. An alternative way to think about this is to say that one must show that

$$\exists P \mid P[A_c \neq A_{c'}] = 0 \forall A_c \sim X_c, A_{c'} \sim X_{c'} \quad (1)$$

where X_c is a random variable in context c , $X_{c'}$ is "the same" random variable in context c' and A_c is a sample from X_c .

Now, while this is all well and good in theory, the question remains—how does this apply to the research we have been doing in quasi-"Agentic-AI" naming games? Simply put, a core aspect of this study is symmetry-breaking in models. This symmetry breaking is where bias comes from, and ultimately it is this bias which is amplified in populations. What is really crucial, however, is that while we can currently answer questions about *how much* bias in any specific circumstance gets amplified, we cannot say anything about generalizability or the ability to predict amplification. Answering the question of contextuality, however, tells *whether or not we can infer the behavior of a model in one context to another*. With regard to our study, this works along two dimensions:

- Syntactic permutations
- Memory configurations.

One thing which we (and others [6]) have noticed is that the output probabilities from models tends to differ greatly based on the order of options presented. For example the prompts, "The plumber left {his, her} tools on the table," renders very different probabilities than, "The plumber left {her, his} tools on the table.". Thus, a first easy but very foundational question that we can ask is whether or not the permutation of gendered pronouns in prompting is contextual. If it is contextual, this leads to serious methodological and testing questions. The most obvious of these being that we *logically* can not extrapolate the behavior of a model about the second context from the first. This also raises questions about whether or not one can simply average the probabilities of the two responses together to understand how a model may behave "out in the wild." In fact, contextuality of this sort would raise questions about our ability to ever sufficiently "lab test" LLMs for gender bias before putting them into production.

In the second axis, we consider the case of memory. Here, the results become somewhat compelling whether or not contextuality exists in this system. In this case, we imagine an agent with some "memory" (say, knowledge of its last interaction included in the prompt) presenting probabilities for "his" and "her" which differ from the agent with no memory. If this system is contextual, then it proves that there is no way that latent variables can explain the emergent phenomena—we cannot predict the collective bias without extracting the probabilities from each context (read 'memory state' or 'prompt') of each model. This too presents significant concerns for the application of models in practical settings. However, if this is *not* a contextual setting, then it is also quite enticing because it would suggest that there can exist some set of latent variables which explain how models adapt based on their 'memory' of past interactions. Finally, there is a possibility in which it is mixed, and some pairs

of measurements are contextual and others aren't. This could occur if we see contextuality in the permutation of pronouns, but not in memory. This would perhaps be the most fascinating, as it could lend a very unexpected insight into the inner workings of these models.

Seeing how this relates to our work, the last question to ask is now: how is this important for mitigating gender bias? Quite simply, this becomes a question of trust and safety in the deployment of language models. If contextuality exists *anywhere*, then it justifies a call for a serious reconsideration of the relationship between testing and deployment of language models. This is because contextuality suggests that while models may not exhibit any gender bias in lab contexts, one would have to assess *all possible* contexts to determine whether the models truly are not misogynistic, transphobic, etc. Moreover, if there is contextuality in memory, this suggests that we may be able to have models that exhibit no bias whatsoever in individual (one-on-one chat) settings, but still see bias emerge in collective settings.

As of now, the research is promising in the sense that there is work already being done on contextuality in LLMs, and results suggest that it is especially present in closely related pairs of words (see [10]).

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